

A Level Computer Science (9618)

Topical Past Paper Worksheet

Topic 1: Information Representation

Subtopic 1.2.1: Multimedia: Graphics

Concept 1.2.1.3: Vector Graphic Encoding (Drawing Object, Property, Drawing List)

Total Questions: 4

Mark Schemes begin on Page 4

Question 1 | Source: 9618_w24_qp_12 (Q7.b)

- 7 A student takes a photograph of a science experiment.
- (b) The student draws a picture on paper that is scanned into the computer and saved as a vector graphic.

Define the vector graphic terms property and drawing list.

Property

.....

Drawing list

.....

[2]

Question 2 | Source: 9618_s24_qp_12 (Q2.d.ii)

- 2 A computer game is being designed that users will be able to play using a virtual reality (VR) headset.
- (d) The computer can transmit a video made from bitmap images and vector graphic animations to the VR headset.
- (ii) Describe the contents of a vector graphic drawing list.

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..... [2]

Question 3 | Source: 9618_w23_qp_11 (Q1.a)

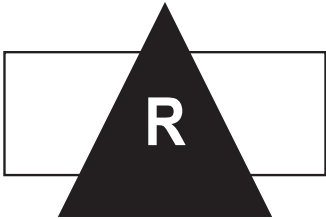
- 1 (a) Draw **one** line from each vector graphic term to its most appropriate description.

Term	Description
drawing list	a component created using a formula
drawing object	defines one characteristic of a component
property	data required to create all components in the graphic

[2]

Question 4 | Source: 9618_w21_qp_12 (Q5.a)

5 Riya has created the following logo as a vector graphic.



(a) Complete the table by writing a description of each vector graphic term **and** give an example for this logo.

Term	Description	Example from logo
Property		
Drawing list		

[4]

MARK SCHEMES

Mark Scheme for Question 1 | Source: 9618_w24_ms_12 (Q7.b)

7(b)	1 mark for each correct definition Property: - an attribute of a drawing object // data about a shape // defines one aspect of the appearance of a drawing object Drawing list: - all the drawing objects/shapes in an image // stores the commands/descriptions / mathematical equations required to draw each object	2
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Mark Scheme for Question 2 | Source: 9618_s24_ms_12 (Q2.d.ii)

2(d)(ii)	1 mark each to max 2: - List of objects in the drawing - A list that stores the command/description/equation required to draw each object - Properties of each object e.g. the fill colour, line weight/colour	2
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Mark Scheme for Question 3 | Source: 9618_w23_ms_11 (Q1.a)

1(a)	2 marks for all 3 lines correct 1 mark for 1 line correct <table><tr><th>Term</th><th>Description</th></tr><tr><td>drawing list</td><td>a component created using a formula</td></tr><tr><td>drawing object</td><td>defines one characteristic of a component</td></tr><tr><td>property</td><td>data required to create all components in the graphic</td></tr></table>	Term	Description	drawing list	a component created using a formula	drawing object	defines one characteristic of a component	property	data required to create all components in the graphic	2
Term	Description									
drawing list	a component created using a formula									
drawing object	defines one characteristic of a component									
property	data required to create all components in the graphic									

Mark Scheme for Question 4 | Source: 9618_w21_ms_12 (Q5.a)

5(a)

1 mark for each description,1 mark for each valid example

4

Term	Description	Example from logo
Property	data about the shapes // defines one aspect of the appearance of the drawing object	e.g. black line // white fill // black fill //solid (line) // font of letter // colour of triangle
Drawing list	the list of shapes involved in an image // a list that stores the command/description required to draw each object	e.g. triangle // capital letter R // rectangle // line